Jiayuan Shen

(240) 756-9786 | jshen20@terpmail.umd.edu | linkedin.com/in/jiayuan-shen-b95352217 | github.com/jiayuan-shen

Research Interests

Natural Language Processing, Machine Learning

Education

University of Maryland — College Park, MD
Ph.D Student in Information Science — *Sep 2026 – Now*

University of Maryland — College Park, MD
M.S. in Applied Machine Learning | GPA: 3.94/4.00 — *Aug 2023 – Dec 2024*

University of Maryland — College Park, MD
B.S. in Information Science | GPA: 4.00/4.00 — *Sep 2020 – Jun 2023*

Research Experience

PACT+AI — LLM-Assisted Preparation for Dyslexia-Friendly Instruction — Jun 2025 – Present
University of Maryland — *College Park, MD*

- Operationalized PACT reading-comprehension steps as coordinated LLM sub-tasks for generating instructional materials.
- Evaluated outputs from prompt ablations against PACT-based checklists to study accessibility and content fidelity.

AIAI Challenge — Multimodal Classification of Instructional Activities — Jun 2025 – Aug 2025
University of Maryland — *College Park, MD*

- Framed classroom-activity recognition as a second-wise multi-label classification task over instructional video with overlapping codes and severe class imbalance.
- Built an training pipeline with imbalance-aware loss that improved macro-F1

Selected Projects

Stable Diffusion Text-to-Image Service on AWS

- Implemented a text-to-image generation pipeline with external conditioning and exposed it as a containerized inference service.
- Automated cloud deployment with infrastructure-as-code, enabling reproducible and scalable experiments with diffusion models.

InfoLlama Query Assistant

- Implemented a Llama-style causal Transformer from scratch and trained it on multi-turn question–answer dialogues as the backbone of a retrieval-augmented query assistant.
- Optimized training and inference with efficient numerical routines and compilation, and analyzed failure modes of small language models trained from scratch.

English–Chinese Transformer Machine Translation

- Built an encoder–decoder Transformer for English–Chinese translation, including data preprocessing, parallel-corpus construction, and auto-regressive decoding.
- Designed training and evaluation with standard optimization and BLEU/BERT Score metrics to study how regularization and model size under limited compute affect translation quality.

Teaching Experience

Undergraduate Teaching Assistant — Sep 2022 – Jun 2023
University of Maryland — *College Park, MD*

- Assisted in Statistics for Information Science and Decision-Making for Information Science; conducted office hours and discussion sessions.
- Mentored students in debugging, model evaluation, and reproducible project workflows.

Honors & Awards

AI for Advancing Instruction (AIAI) Challenge — 3rd Place (DrivenData) — Aug 2024
Summa Cum Laude — B.S. in Information Science, University of Maryland — Jun 2023

Technical Skills

Languages: Python, SQL, R, HTML/CSS, JavaScript, C/C++, Bash
Developer Tools: Linux, VS Code, Git, Docker, Amazon Web Service, PostgreSQL, MySQL, Tableau, OpenRefine
Libraries: PyTorch, Hugging Face Transformers, NumPy, Pandas, scikit-learn, Pydantic, Matplotlib